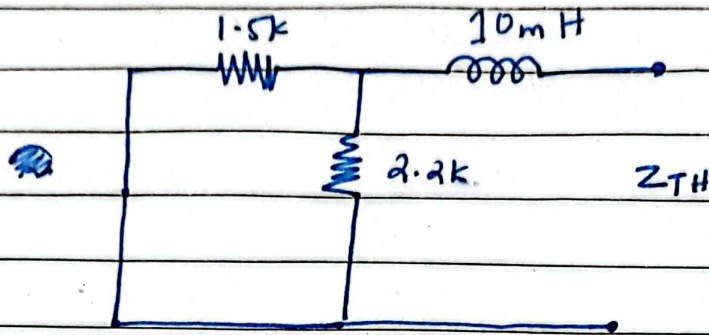


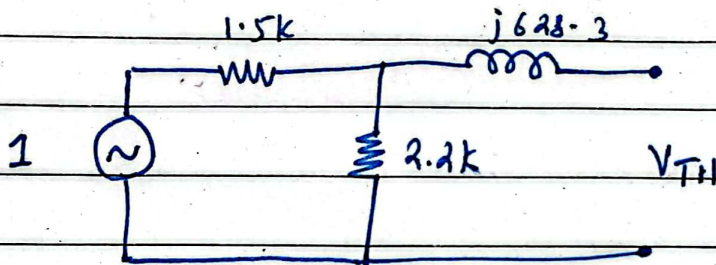
Prelab 8

Q.1)



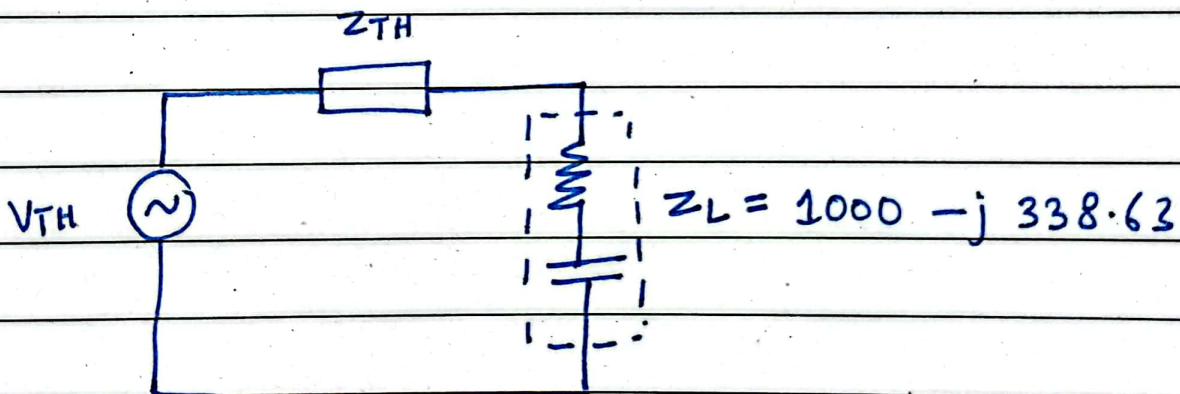
$$Z_{TH} = (1.5k \parallel 2.2k) + j(2\pi(10k)(10 \times 10^{-3}))$$

$$Z_{TH} = 891.9 + j628.3 \quad | \quad 1090.9 / 35.16^\circ$$



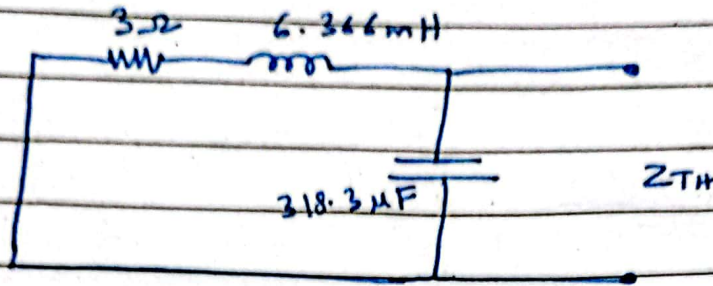
$$V_{TH} = \left(\frac{2.2k}{1.5k + 2.2k} \right) (1 \angle 0^\circ)$$

$$V_{TH} = 0.5946V$$



$$V_{load} = \left(\frac{1000 - j338.63}{Z_{TH}} \right) \quad | \quad V_{load} = 0.328 \angle -27.4^\circ$$

Q.2)

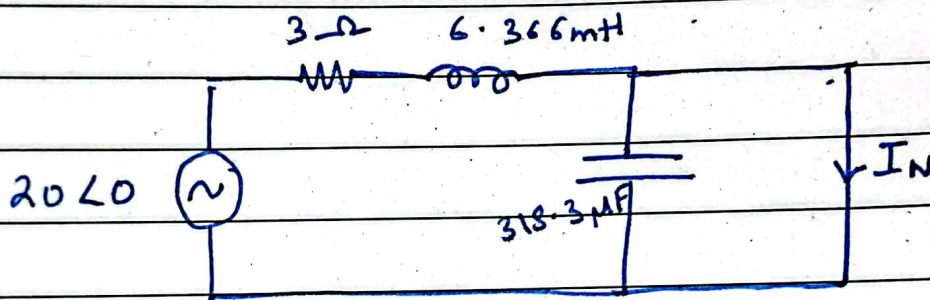


$$Z_{TH} = (3 + j\omega L) \parallel 1/j\omega C$$

$$= (3 + j3.999) \parallel -j5$$

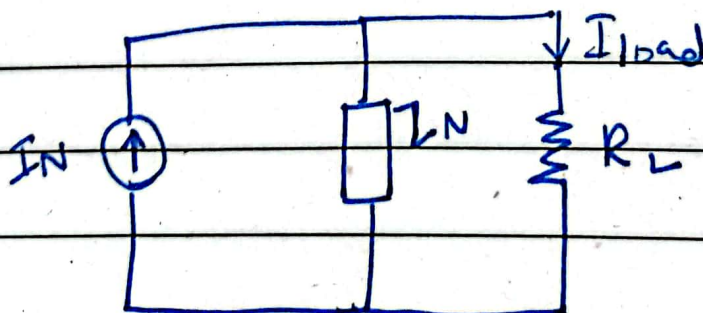
$$Z_{TH} = 7.4997 - j2.4997$$

$$Z_{TH} = 7.905 \angle -18.313$$



$$I_N = \frac{20 \angle 0}{3 + j3.999} = 2.4008 - j3.2$$

$$I_N = 4.00 \angle -53.123$$



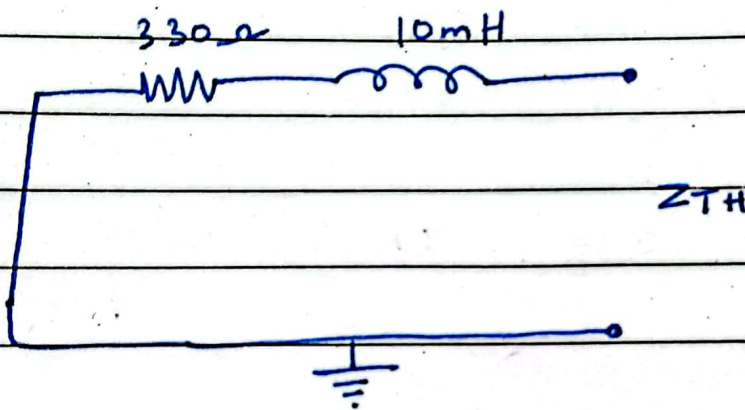
$$I_{load} = \frac{Z_N}{R_L + Z_N} \times I_N$$

$$R_L = 10 \Omega$$

$$= \left(\frac{7.905 \angle -18.43}{10 + 7.905 \angle -18.43} \right) (4 \angle -53.123)$$

$$I_{Load} = 0.8 - j1.5997 = \boxed{1.789 \angle -63.43}$$

Q.3)



$$Z_{TH} = 330 + j(2\pi(5000)(10 \times 10^{-3}))$$

$$\boxed{Z_{TH} = 330 + j314.16}$$

$$\boxed{Z_{LOAD} = 330 - j314.16}$$